Statistics/Data analysis

```
      name:  <unnamed>
       log:  E:\Documents\STATA\cholesterol part 2.smcl
  log type:  smcl
 opened on:  11 May 2026, 16:49:12

1 .
2 . *import data
3 . import delimited "E:\Documents\STATA\assignment 2_Number 2_2025_2026.csv", clear
  (encoding automatically selected: ISO-8859-2)
  (8 vars, 4,699 obs)

4 .
5 . *RENAME COLUMNS
6 . rename column1 sex

7 . rename column2 sbp

8 . rename column3 dbp

9 . rename column4 chol

10. rename column5 age

11. rename column8 bmi

12.
13. label define sexlbl 1 "Male" 2 "Female"

14. label values sex sexlbl

15.
16. * Generate ordinal outcome
17. gen chol_cat = .
  (4,699 missing values generated)

18. replace chol_cat = 0 if chol < 200
  (1,237 real changes made)

19. replace chol_cat = 1 if chol >= 200 & chol < 240
  (1,752 real changes made)

20. replace chol_cat = 2 if chol >= 240
  (1,710 real changes made)

21.
22. label define chollbl 0 "Healthy" 1 "Borderline High" 2 "High"

23. label values chol_cat chollbl

24.
25. * Verify
26. tab chol_cat, missing
```

chol_cat	Freq.	Percent	Cum.
Healthy	**1,237**	**26.32**	**26.32**
Borderline High	**1,752**	**37.28**	**63.61**
High	**1,710**	**36.39**	**100.00**
Total	**4,699**	**100.00**	

```
27. summarize sbp dbp age bmi
```

Variable	Obs	Mean	Std. dev.	Min	Max
sbp	**4,699**	**132.7665**	**22.8011**	**80**	**270**
dbp	**4,699**	**82.5416**	**12.73732**	**40**	**148**
age	**4,699**	**46.04107**	**8.504363**	**30**	**68**
bmi	**4,690**	**25.63171**	**4.094672**	**16.2**	**57.6**

```
28. tab sex
```

Column 1	Freq.	Percent	Cum.
Male	**2,049**	**43.61**	**43.61**
Female	**2,650**	**56.39**	**100.00**
Total	**4,699**	**100.00**	

```
29.
30.
31. *COLLINEARITY CHECK
32.
33.
34. * 2a. Fit full model
35. ologit chol_cat sex sbp dbp age bmi, nolog
```

Ordered logistic regression — Number of obs = **4,690**; LR chi2(**5**) = **454.55**; Prob > chi2 = **0.0000**

Log likelihood = **-4871.077**; Pseudo R2 = **0.0446**

chol_cat	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
sex	**.0046155**	**.055908**	**0.08**	**0.934**	**-.1049621**	**.1141931**
sbp	**.0035541**	**.0020854**	**1.70**	**0.088**	**-.0005332**	**.0076414**
dbp	**.0093427**	**.0035914**	**2.60**	**0.009**	**.0023036**	**.0163817**
age	**.0527691**	**.0036321**	**14.53**	**0.000**	**.0456503**	**.0598879**
bmi	**.0377099**	**.0074459**	**5.06**	**0.000**	**.0231162**	**.0523035**
/cut1	**3.538727**	**.2641314**			**3.021039**	**4.056415**
/cut2	**5.255886**	**.2707671**			**4.725193**	**5.78658**

```
36.
37. * 2b. Predict the linear predictor
38.
39. predict logits, xb
  (9 missing values generated)

40.
41. * 2c. Run linear regression of logits on predictors
42. regress logits sex sbp dbp age bmi
```

Source	SS	df	MS		
Model	**1701.09649**	**5**	**340.219297**	Number of obs	= **4,690**
Residual	**7.8575e-11**	**4,684**	**1.6775e-14**	F(5, 4684)	> **99999.00**
Total	**1701.09649**	**4,689**	**.362784493**	Prob > F	= **0.0000**
				R-squared	= **1.0000**
				Adj R-squared	= **1.0000**
				Root MSE	= **1.3e-07**

logits	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
sex	**.0046155**	**3.86e-09**	**1.2e+06**	**0.000**	**.0046155**	**.0046155**
sbp	**.0035541**	**1.43e-10**	**2.5e+07**	**0.000**	**.0035541**	**.0035541**
dbp	**.0093427**	**2.47e-10**	**3.8e+07**	**0.000**	**.0093427**	**.0093427**
age	**.0527691**	**2.44e-10**	**2.2e+08**	**0.000**	**.0527691**	**.0527691**
bmi	**.0377099**	**5.00e-10**	**7.5e+07**	**0.000**	**.0377099**	**.0377099**
_cons	**-1.50e-08**	**1.76e-08**	**-0.85**	**0.394**	**-4.95e-08**	**1.95e-08**

```
43.
44. estat vif
```

Variable	VIF	1/VIF
sbp	**2.96**	**0.338381**
dbp	**2.77**	**0.361595**
age	**1.21**	**0.829413**
bmi	**1.17**	**0.852495**
sex	**1.02**	**0.977648**
Mean VIF	**1.82**	

```
45.
46. *No collinearity since no vif >10
47.
48.
49. *VARIABLE SELECTION
50. *since we have large sample size; we form all possible interaction terms
51.
52.
53. * ASSESS INTERACTION
54. *full model;
55. ologit chol_cat i.sex##c.sbp i.sex##c.dbp i.sex##c.age i.sex##c.bmi c.sbp##c.dbp c.s
  > bp##c.age c.sbp##c.bmi c.dbp##c.age c.dbp##c.bmi c.age##c.bmi, nolog or
  note: sbp omitted because of collinearity.
  note: dbp omitted because of collinearity.
  note: sbp omitted because of collinearity.
  note: age omitted because of collinearity.
  note: sbp omitted because of collinearity.
  note: bmi omitted because of collinearity.
  note: dbp omitted because of collinearity.
  note: age omitted because of collinearity.
  note: dbp omitted because of collinearity.
  note: bmi omitted because of collinearity.
  note: age omitted because of collinearity.
  note: bmi omitted because of collinearity.

  Ordered logistic regression                       Number of obs =  4,690
                                                    LR chi2(15)   = 647.81
                                                    Prob > chi2   = 0.0000
  Log likelihood = -4774.4428                       Pseudo R2     = 0.0635
```

chol_cat	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
sex						
Female	**.1397815**	**.0732782**	**-3.75**	**0.000**	**.0500289**	**.3905518**
sbp	**1.02396**	**.0170735**	**1.42**	**0.156**	**.9910375**	**1.057976**
sex#c.sbp						
Female	**.993355**	**.0043009**	**-1.54**	**0.124**	**.9849611**	**1.00182**
dbp	**1.058633**	**.0290272**	**2.08**	**0.038**	**1.003243**	**1.117082**
sex#c.dbp						
Female	**1.010172**	**.0074082**	**1.38**	**0.168**	**.995756**	**1.024797**
age	**1.169836**	**.0358795**	**5.11**	**0.000**	**1.101585**	**1.242315**
sex#c.age						
Female	**1.083094**	**.0080169**	**10.78**	**0.000**	**1.067495**	**1.098921**
bmi	**1.209871**	**.0688808**	**3.35**	**0.001**	**1.082127**	**1.352695**
sex#c.bmi						
Female	**.9401549**	**.0151717**	**-3.82**	**0.000**	**.9108842**	**.970366**
sbp	**1**	(omitted)				
dbp	**1**	(omitted)				
c.sbp#c.dbp	**.9998711**	**.0000759**	**-1.70**	**0.089**	**.9997224**	**1.00002**

sbp	1	(omitted)				
age	1	(omitted)				
c.sbp#c.age	.9999729	.0002486	-0.11	0.913	.9994857	1.00046
sbp	1	(omitted)				
bmi	1	(omitted)				
c.sbp#c.bmi	.9998643	.0004763	-0.28	0.776	.9989311	1.000798
dbp	1	(omitted)				
age	1	(omitted)				
c.dbp#c.age	.9991	.0004425	-2.03	0.042	.9982331	.9999676
dbp	1	(omitted)				
bmi	1	(omitted)				
c.dbp#c.bmi	1.000205	.0008503	0.24	0.809	.9985401	1.001873
age	1	(omitted)				
bmi	1	(omitted)				
c.age#c.bmi	.9974049	.0009525	-2.72	0.007	.9955398	.9992734
/cut1	10.70392	1.593106			7.581489	13.82635
/cut2	12.47599	1.59622			9.347454	15.60452

Note: Estimates are transformed only in the first equation to odds ratios.

```
56. estimates store full

57.
58.
59. *reduced model
60. ologit chol_cat sex sbp dbp age bmi, nolog
```

Ordered logistic regression — Number of obs = **4,690**
LR chi2(**5**) = **454.55**
Prob > chi2 = **0.0000**
Log likelihood = **-4871.077** — Pseudo R2 = **0.0446**

chol_cat	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
sex	.0046155	.055908	0.08	0.934	-.1049621	.1141931
sbp	.0035541	.0020854	1.70	0.088	-.0005332	.0076414
dbp	.0093427	.0035914	2.60	0.009	.0023036	.0163817
age	.0527691	.0036321	14.53	0.000	.0456503	.0598879
bmi	.0377099	.0074459	5.06	0.000	.0231162	.0523035
/cut1	3.538727	.2641314			3.021039	4.056415
/cut2	5.255886	.2707671			4.725193	5.78658

```
61. estimates store reduced

62.
63. *compare
```

```
64. lrtest reduced full, stats

  Likelihood-ratio test
  Assumption: reduced nested within full

  LR chi2(10) = 193.27
  Prob > chi2 = 0.0000

  Akaike's information criterion and Bayesian information criterion
```

Model	N	ll(null)	ll(model)	df	AIC	BIC
reduced	4,690	-5098.35	-4871.077	7	9756.154	9801.326
full	4,690	-5098.35	-4774.443	17	9582.886	9692.59

Note: BIC uses N = number of observations. See **[R] BIC note**.

```
65.
66. *significant pvalue means presence of interaction
67.
68. *use step wise to drop nonsignificant terms
69. sw, pr(0.05) lr: ologit chol_cat sex sbp dbp age bmi i.sex#c.sbp i.sex#c.dbp i.sex#c
  > .age i.sex#c.bmi c.sbp#c.dbp c.sbp#c.age c.sbp#c.bmi c.dbp#c.age c.dbp#c.bmi c.age#c
  > .bmi
  note: 2.sex#c.sbp omitted because of estimability.
  note: 2.sex#c.dbp omitted because of estimability.
  note: 2.sex#c.age omitted because of estimability.
  note: 2.sex#c.bmi omitted because of estimability.

  LR test, begin with full model:
  p = 0.9131 >= 0.0500, removing c.sbp#c.age
  p = 0.7947 >= 0.0500, removing c.dbp#c.bmi
  p = 0.8707 >= 0.0500, removing c.sbp#c.bmi
  p = 0.1571 >= 0.0500, removing 1.sex#c.dbp
  p = 0.4383 >= 0.0500, removing 1.sex#c.sbp
  p = 0.0529 >= 0.0500, removing c.sbp#c.dbp
  p = 0.1281 >= 0.0500, removing sbp

  Ordered logistic regression                     Number of obs =  4,690
                                                  LR chi2(8)    =  639.04
                                                  Prob > chi2   =  0.0000
  Log likelihood = -4778.8292                     Pseudo R2     =  0.0627
```

chol_cat	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
sex	-2.004408	.4711707	-4.25	0.000	-2.927886	-1.080931
sex#c.bmi						
Male	.0603401	.0150861	4.00	0.000	.0307719	.0899082
dbp	.0617573	.0138585	4.46	0.000	.0345951	.0889195
age	.2497441	.0269193	9.28	0.000	.1969833	.3025049
bmi	.1324404	.041286	3.21	0.001	.0515215	.2133594
c.age#c.bmi	-.0026693	.0008526	-3.13	0.002	-.0043403	-.0009984
c.dbp#c.age	-.0010653	.000285	-3.74	0.000	-.0016239	-.0005068
sex#c.age						
Male	-.0788897	.0068865	-11.46	0.000	-.092387	-.0653923
/cut1	7.586077	1.482156			4.681105	10.49105
/cut2	9.355467	1.484728			6.445453	12.26548

```
70.
71. *Model with significant interactio terms is
72. ologit chol_cat i.sex i.sex#c.bmi dbp age bmi c.age#c.bmi  c.dbp#c.age i.sex#c.age,
  > nolog

Ordered logistic regression                       Number of obs =  4,690
                                                  LR chi2(8)    =  639.04
                                                  Prob > chi2   =  0.0000
Log likelihood = -4778.8292                       Pseudo R2     =  0.0627

------------------------------------------------------------------------------
    chol_cat | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         sex |
      Female |  -2.004408   .4711707    -4.25   0.000    -2.927886   -1.080931
             |
     sex#c.bmi |
      Female |  -.0603401   .0150861    -4.00   0.000    -.0899082   -.0307719
             |
         dbp |   .0617573   .0138585     4.46   0.000     .0345951    .0889195
         age |   .1708544   .0272645     6.27   0.000     .1174171    .2242918
         bmi |   .1927805   .0413053     4.67   0.000     .1118237    .2737374
             |
 c.age#c.bmi |  -.0026693   .0008526    -3.13   0.002    -.0043403   -.0009984
             |
 c.dbp#c.age |  -.0010653    .000285    -3.74   0.000    -.0016239   -.0005068
             |
     sex#c.age |
      Female |   .0788897   .0068865    11.46   0.000     .0653923     .092387
-------------+----------------------------------------------------------------
       /cut1 |   9.590485   1.295531                      7.05129    12.12968
       /cut2 |   11.35987   1.298968                      8.813944   13.90581
------------------------------------------------------------------------------

73.
74.
75. *ASSESS CONFOUNDING
76. *bring back dropped variable sbp and check for confounding;
77. *before sbp
78. ologit chol_cat i.sex i.sex#c.bmi dbp age bmi c.age#c.bmi  c.dbp#c.age i.sex#c.age,
  > nolog or

Ordered logistic regression                       Number of obs =  4,690
                                                  LR chi2(8)    =  639.04
                                                  Prob > chi2   =  0.0000
Log likelihood = -4778.8292                       Pseudo R2     =  0.0627

------------------------------------------------------------------------------
    chol_cat | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         sex |
      Female |     .13474   .0634856    -4.25   0.000     .0535101    .3392797
             |
     sex#c.bmi |
      Female |   .9414443   .0142027    -4.00   0.000     .9140151    .9696967
             |
         dbp |   1.063704   .0147414     4.46   0.000       1.0352    1.092993
         age |   1.186318   .0323443     6.27   0.000     1.124588    1.251436
         bmi |   1.212617   .0500875     4.67   0.000     1.118316    1.314869
             |
 c.age#c.bmi |   .9973342   .0008503    -3.13   0.002     .9956691    .9990021
             |
 c.dbp#c.age |   .9989352   .0002847    -3.74   0.000     .9983774    .9994934
             |
     sex#c.age |
      Female |   1.082085   .0074518    11.46   0.000     1.067578    1.096789
-------------+----------------------------------------------------------------
       /cut1 |   9.590485   1.295531                      7.05129    12.12968
       /cut2 |   11.35987   1.298968                      8.813944   13.90581
------------------------------------------------------------------------------
Note: Estimates are transformed only in the first equation to odds ratios.
```

```
79.
80. *after adding sbp
81.
82. ologit chol_cat i.sex i.sex#c.bmi dbp sbp age bmi c.age#c.bmi  c.dbp#c.age i.sex#c.a
  > ge, nolog or

Ordered logistic regression                       Number of obs =  4,690
                                                  LR chi2(9)    = 641.36
                                                  Prob > chi2   = 0.0000
Log likelihood = -4777.6714                       Pseudo R2     = 0.0629

------------------------------------------------------------------------------
    chol_cat | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         sex |
      Female |   .1461816   .0693187    -4.06   0.000     .0577107    .3702791
             |
   sex#c.bmi |
      Female |   .9404314   .0142093    -4.06   0.000     .9129899    .9686977
             |
         dbp |   1.062805   .0147473     4.39   0.000      1.03429    1.092106
         sbp |   1.003283   .0021634     1.52   0.128     .9990519    1.007532
         age |   1.193179    .032853     6.41   0.000     1.130495    1.259339
         bmi |    1.21518   .0502645     4.71   0.000     1.120551      1.3178
             |
 c.age#c.bmi |   .9972936   .0008514    -3.17   0.002     .9956262    .9989637
             |
 c.dbp#c.age |   .9988649   .0002886    -3.93   0.000     .9982994    .9994307
             |
   sex#c.age |
      Female |   1.080634   .0075011    11.17   0.000     1.066032    1.095437
-------------+----------------------------------------------------------------
       /cut1 |    9.95245   1.317547                      7.370105     12.5348
       /cut2 |    11.7225   1.321036                      9.133315    14.31168
------------------------------------------------------------------------------
Note: Estimates are transformed only in the first equation to odds ratios.

83.
84.
85. *sbp is not a confounder;
86.
87. *MODEL AFTER CONFOUNDING ASSESSMENT
88. ologit chol_cat i.sex i.sex#c.bmi dbp age bmi c.age#c.bmi  c.dbp#c.age i.sex#c.age,
  > nolog or

Ordered logistic regression                       Number of obs =  4,690
                                                  LR chi2(8)    = 639.04
                                                  Prob > chi2   = 0.0000
Log likelihood = -4778.8292                       Pseudo R2     = 0.0627

------------------------------------------------------------------------------
    chol_cat | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         sex |
      Female |     .13474   .0634856    -4.25   0.000     .0535101    .3392797
             |
   sex#c.bmi |
      Female |   .9414443   .0142027    -4.00   0.000     .9140151    .9696967
             |
         dbp |   1.063704   .0147414     4.46   0.000       1.0352    1.092993
         age |   1.186318   .0323443     6.27   0.000     1.124588    1.251436
         bmi |   1.212617   .0500875     4.67   0.000     1.118316    1.314869
             |
 c.age#c.bmi |   .9973342   .0008503    -3.13   0.002     .9956691    .9990021
             |
 c.dbp#c.age |   .9989352   .0002847    -3.74   0.000     .9983774    .9994934
             |
   sex#c.age |
      Female |   1.082085   .0074518    11.46   0.000     1.067578    1.096789
-------------+----------------------------------------------------------------
       /cut1 |   9.590485   1.295531                      7.05129    12.12968
       /cut2 |   11.35987   1.298968                     8.813944    13.90581
------------------------------------------------------------------------------
```

Note: Estimates are transformed only in the first equation to odds ratios.

```
89.
90.
91.
92.
93. *FINAL MODEL AND GOODNESS OF FIT
94. ologit chol_cat i.sex i.sex#c.bmi dbp age bmi c.age#c.bmi  c.dbp#c.age i.sex#c.age,
  > nolog or
```

Ordered logistic regression — Number of obs = **4,690**; LR chi2(**8**) = **639.04**; Prob > chi2 = **0.0000**; Log likelihood = **-4778.8292**; Pseudo R2 = **0.0627**

chol_cat	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
sex						
Female	**.13474**	**.0634856**	**-4.25**	**0.000**	**.0535101**	**.3392797**
sex#c.bmi						
Female	**.9414443**	**.0142027**	**-4.00**	**0.000**	**.9140151**	**.9696967**
dbp	**1.063704**	**.0147414**	**4.46**	**0.000**	**1.0352**	**1.092993**
age	**1.186318**	**.0323443**	**6.27**	**0.000**	**1.124588**	**1.251436**
bmi	**1.212617**	**.0500875**	**4.67**	**0.000**	**1.118316**	**1.314869**
c.age#c.bmi	**.9973342**	**.0008503**	**-3.13**	**0.002**	**.9956691**	**.9990021**
c.dbp#c.age	**.9989352**	**.0002847**	**-3.74**	**0.000**	**.9983774**	**.9994934**
sex#c.age						
Female	**1.082085**	**.0074518**	**11.46**	**0.000**	**1.067578**	**1.096789**
/cut1	**9.590485**	**1.295531**			**7.05129**	**12.12968**
/cut2	**11.35987**	**1.298968**			**8.813944**	**13.90581**

Note: Estimates are transformed only in the first equation to odds ratios.

```
95.
96.
97. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	**4,690**	**-5098.35**	**-4778.829**	**10**	**9577.658**	**9642.19**

Note: BIC uses N = number of observations. See **[R] BIC note**.

```
98.
99. estat vce
```

Covariance matrix of coefficients of **ologit** model

	chol_cat						
	/						
	2.sex	2.sex#c.bmi	dbp	age	bmi	c.age#c.bmi	c.dbp#c.age
	cut1	cut2					
chol_cat							
2.sex	**.22200179**						
2.sex#c.bmi	**-.00522695**	**.00022759**					

```
>                        |
         dbp |   .00015849   5.038e-06   .00019206
>                        |
         age |    .0009689   .00003046   .00022422   .00074335
>                        |
         bmi |   .00327251  -.00011459  -.00019089   .00055762   .00170613
>                        |
 c.age#c.bmi |   .00001136  -8.643e-07   3.720e-06  -.00001258  -.00003351   7.268e-07
>                        |
 c.dbp#c.age |  -2.539e-06  -8.313e-08  -3.889e-06  -4.746e-06   3.770e-06  -7.595e-08
>   8.122e-08            |
 2.sex#c.age |  -.00183579   -.0000135  -4.668e-06  -.00003306  -2.030e-06   1.342e-07
>   7.101e-08   .00004742 |
-------------------------+------------------------------------------------------------
                         |
/                        |
>                        |
       cut1 |   .14819111  -.00248342   .01084265   .03402597   .02843207   -.0005616
>   -.0002228   -.0015412 |  1.6784018
       cut2 |   .14754911  -.00251106   .01086893   .03409699   .02851279  -.00056265
>  -.00022324  -.00151161 |  1.6821482   1.6873183

100
101
102 ologit chol_cat i.sex i.sex#c.bmi dbp age bmi c.age#c.bmi  c.dbp#c.age i.sex#c.age,
  > nolog

Ordered logistic regression                     Number of obs =  4,690
                                                LR chi2(8)    =  639.04
                                                Prob > chi2   =  0.0000
Log likelihood = -4778.8292                     Pseudo R2     =  0.0627

------------------------------------------------------------------------------
    chol_cat | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         sex |
      Female |  -2.004408   .4711707    -4.25   0.000    -2.927886   -1.080931
             |
   sex#c.bmi |
      Female |  -.0603401   .0150861    -4.00   0.000    -.0899082   -.0307719
             |
         dbp |   .0617573   .0138585     4.46   0.000     .0345951    .0889195
         age |   .1708544   .0272645     6.27   0.000     .1174171    .2242918
         bmi |   .1927805   .0413053     4.67   0.000     .1118237    .2737374
             |
 c.age#c.bmi |  -.0026693   .0008526    -3.13   0.002    -.0043403   -.0009984
             |
 c.dbp#c.age |  -.0010653    .000285    -3.74   0.000    -.0016239   -.0005068
             |
   sex#c.age |
      Female |   .0788897   .0068865    11.46   0.000     .0653923     .092387
-------------+----------------------------------------------------------------
       /cut1 |   9.590485   1.295531                      7.05129    12.12968
       /cut2 |   11.35987   1.298968                     8.813944    13.90581
------------------------------------------------------------------------------

103
104
105 log close
       name:  <unnamed>
        log:  E:\Documents\STATA\cholesterol part 2.smcl
   log type:  smcl
  closed on:  11 May 2026, 16:49:23
```